

Chrome Spinels from the Jinchuan Ni-Cu Sulfide Deposit, Gansu Province, People's Republic of China

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Abstract

Chrome spinels occur in olivine-rich cumulates in the Jinchuan intrusion, host to one of the largest known accumulations of magmatic Ni-Cu in the world. The chemistry and mode of occurrence of these spinels is compared with those in two other similar but unmineralized mafic-ultramafic intrusions in the same belt.

The Jinchuan intrusion contains Mg-poor chromites with widely varying TiO₂ contents, showing many features that are typical of chromites in normal tholeiitic layered intrusions. Wide compositional variance occurs on the scale of single thin sections, and is attributed to reaction between cumulus chromite and trapped inter-cumulus liquid. This process operated extensively, even on grains that were armored by cumulus olivine crystals. Different reaction paths correlate with the abundance of sulfides in the rock.

The unmineralized intrusions show similar reaction trends, but are offset from the Jinchuan trends. Yejili spinels are generally higher in Al and Ga, while Zangbutai chromites are systematically higher in Fe³⁺. Jinchuan spinel-olivine pairs record equilibration temperatures well below the solidus, and are slightly but significantly lower than blocking temperatures recorded at Yejili.

The Jinchuan ore zone contains low Cr, high Fe³⁺ chromites that are unusual and distinctive. A plausible model for these unusual compositions is that these grains originated as normal aluminous spinels crystallizing from the parent silicate melt, and subsequently became enlarged and modified during crystallization of the sulfide melt. Ti-enriched chromites enclosed within sulfides record a prehistory of extensive reaction with trapped liquid.

Sulfide-associated chromites from Jinchuan are strongly nickel depleted relative to expected Ni values for their Fe³⁺ content when compared with the barren intrusions. This feature, combined with the distinctive chemistry of the sulfide-related spinels, may have applications in the use of heavy resistate minerals during exploration.

Distinctively high Ti chromites are also a feature of intrusions associated with the Karoo flood basalts, suggesting that Jinchuan may have a similar affinity. However, TiO₂ enrichment in chromite can be a product of interaction with differentiated trapped liquid, as indicated by similar trends in lava lakes, layered intrusions, and at Sudbury, and therefore cannot be taken to prove unusually Ti-rich parent magmas.

Introduction

THE JINCHUAN Ni-Cu sulfide deposit is one of the largest in the world, and one of the most enigmatic. As much as half a billion tonnes of disseminated Ni-Cu sulfide ore is hosted by a relatively small intrusive body of dominantly olivine-rich ultramafic rocks, probable high MgO basaltic parentage (Chai and Naldrett, 1992a, b), and with no adjacent external source of sulfur. Chrome spinels are widespread accessory minerals throughout the intrusion. The purpose of this study is to determine what can be learned about the genesis and crystallization history of the intrusion from the chemistry of these spinels.

Samples were analyzed from the Jinchuan intrusion itself, and from two other nearby ultramafic intrusions of similar composition, geometry, and tectonic association, the Yejili and Zangbutai bodies (Fig. 1; Tang, 1991; Tang et al., 1992a).

Geological Setting

The geology of the Jinchuan intrusion has been described by Tang and coworkers (Tang, 1991; Tang et al., 1992a) and

Chai and Naldrett (1992a, b). The intrusion is one of a series of similar mafic-ultramafic bodies located in an uplifted, northwest-striking terrane of early Proterozoic migmatites, biotite gneisses, and marbles that are bounded on both sides by regional faults at the southwest margin of the Sino-Korean platform. The intrusions were emplaced along these regional faults, and are broadly dike-like in morphology.

The Jinchuan intrusion

Geology: The Jinchuan intrusion, which hosts more than 500 million tonnes (Mt) of ore grading 1.2 percent Ni and 0.7 percent Cu, is a northwest-elongated body of ultramafic rocks 6,000 m long and 300 m wide. It was intruded into a Middle Proterozoic (Changchengian) sequence of marbles and granitoids, and has been dated at 1508 ± 31 Ma (Tang et al., 1992b). Chai and Naldrett (1992a, b) interpreted it as the lower eroded portion of a layered intrusion, representing the deep level of a funnel-shaped, linear dike of similar scale to the Great Dyke of Zimbabwe. The intrusion crystallized from a moderately high Mg basaltic magma, and has a central core of dunite and olivine-rich peridotite flanked by ilmenite and olivine websterite. It is divided into three subchambers along its length that are separated by steep, crosscutting

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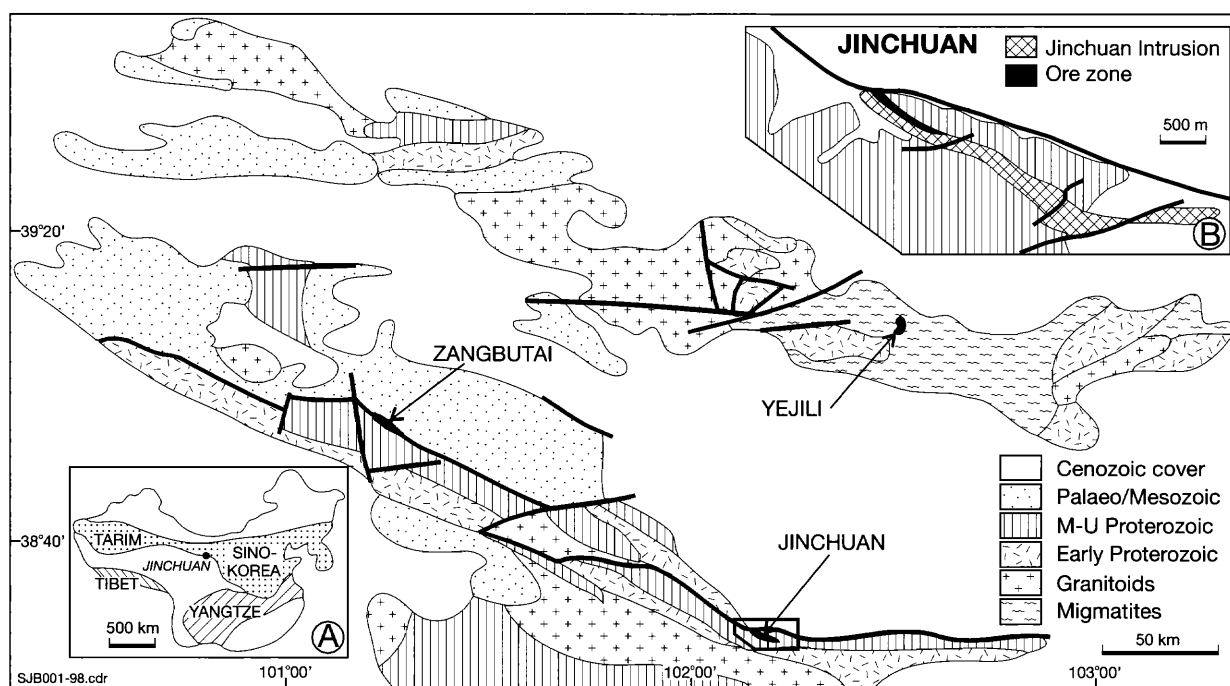


FIG. 1. Regional geological map of the Jinchuan region, Gansu province, north-central China, after Tang (1991). Inset shows geology of the Jinchuan intrusion after Chai and Naldrett (1992b). Granitoids and migmatites are of Early Proterozoic age. Middle Upper Proterozoic (Changchengian) group is composed of mainly carbonate rocks, now marbles, in Jinchuan region. The Early Proterozoic (Changchengian) Group consists of a calc-alkalic greenstone sequence. Granites and migmatites are Early Proterozoic, pre-Jinchuan, in age.

faults. Samples for this study came from three regions of the intrusion:

1. Samples from the main underground number 1 orebody in the west-central subchamber are designated as "Jinchuan ore zone" samples.
2. Samples from the Longzhou open pit, containing mainly low-grade (<1% Ni) disseminated sulfides in the western subchamber, are designated as "Jinchuan disseminated" samples.
3. Samples from a surface traverse across the intrusion, above the number 1 orebody, contain little or no sulfide and are designated as "Jinchuan S-poor" samples. This traverse passes from websterite on the southern margin through sulfide-poor peridotites in the core to ilherzolites on the northern margin.

The olivine-rich rocks that host the spinels and the ore are typical medium- to coarse-grained olivine orthocumulates, containing 60 to 80 percent original olivine with variably altered orthopyroxene, plagioclase, and clinopyroxene (in decreasing order of abundance) as interstitial and poikilitic phases (Fig. 2A, B, C). Olivine in all samples is between about 50 and 80 percent serpentinized, with abundant relic mesh cores of fresh olivine in mesh-textured lizardite (Fig. 2A, B, C). Chlorite is locally abundant as a replacement of interstitial phases. The development of chlorite accompanied partial replacement of chromite by magnetite.

The dominant rock type in the main ore zone itself is net-textured or matrix ore, consisting of partially serpentinized olivine in a continuous matrix of pyrrhotite, pentlandite, and

chalcopyrite (Fig. 2D, E, F), a typical texture for magmatic sulfide ores. In disseminated ores, sulfide aggregates form isolated interstitial blebs between olivine grains. There is no visible secondary violarite, marcasite, or pyrite in the samples studied, implying little or no supergene alteration.

Oxide minerals: The Jinchuan intrusion contains a complex assemblage of oxide minerals, including chromite, titanite magnetite, chromian magnetite, ilmenite, and rare hematite. Chromite is restricted to rocks containing cumulus olivine, and is missing from the marginal websterite unit, which contains Ti magnetite as the only primary oxide phase.

Chromite is the most common oxide mineral, and occurs mostly as small (30 to 200 μm) subhedral to euhedral grains, partially or entirely enclosed in partially serpentinized cumulus olivine (Fig. 2), and for the most part free of magnetite rims. Commonly, as shown in Fig. 2A, chromites are slightly more fine grained in the center of olivine grains than at the margins. Chromites that are totally enclosed in olivine have a median maximum apparent diameter of 70 μm (standard deviation of 50) compared with 150 μm (standard deviation of 50) for all other chromites. This suggests that chromite and olivine have grown simultaneously from the magma. Centrally located chromite grains have been overtaken and trapped by growth of adjacent olivine grains at an early stage, while other grains now at the edge of olivine grains were able to grow to larger sizes before being trapped. Chromite also less commonly occurs enclosed within intercumulus pyroxenes, or less commonly still, within interstitial amphibole or plagioclase grains. Chromites outside of olivine grains are typically less euhedral and are relatively coarse (Fig. 2C).

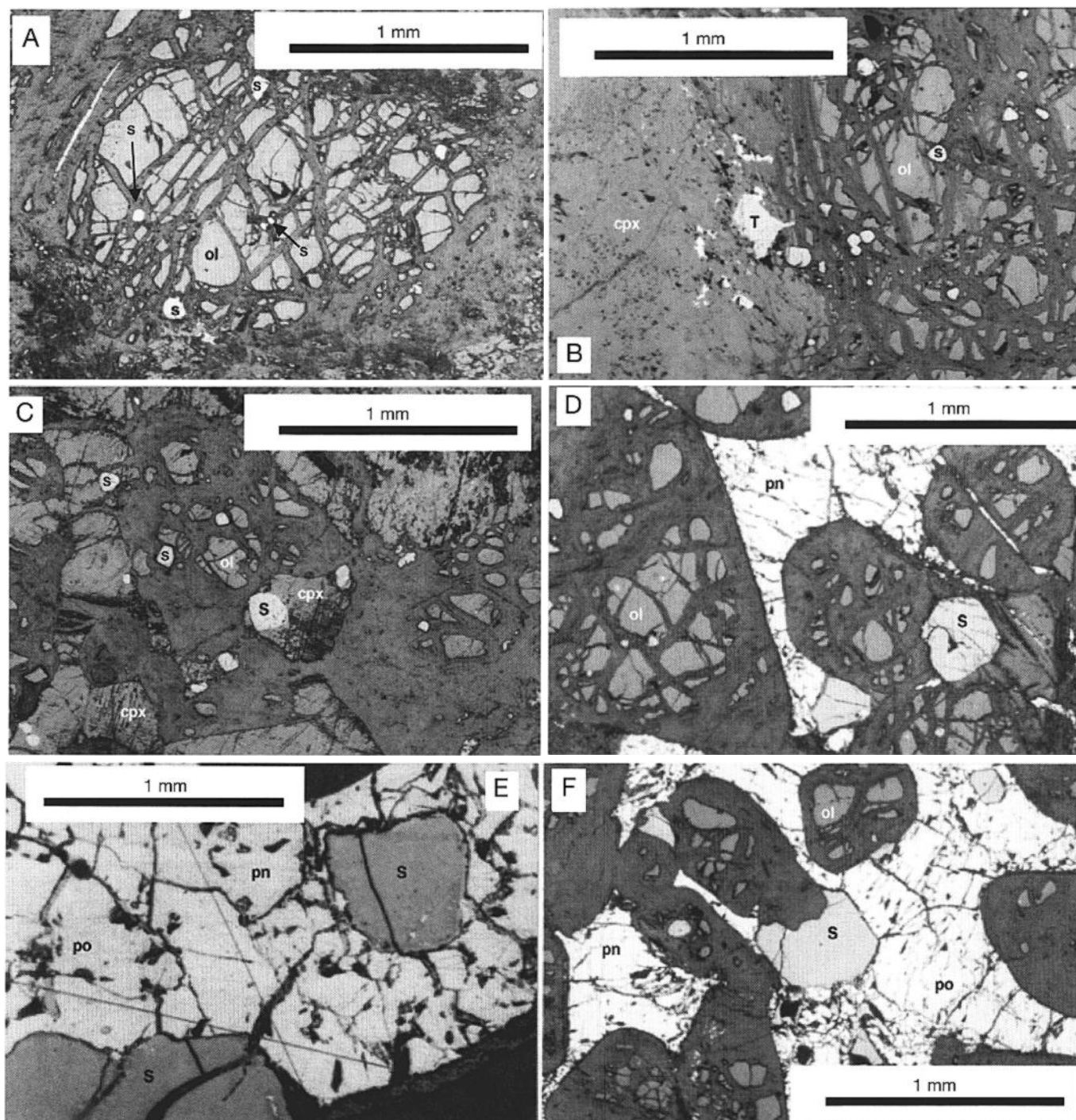


FIG. 2. Photomicrographs of chrome spinels in olivine cumulates of the Jinchuan intrusion. A. Chromite (light gray, small euhedral grains (s)), enclosed in single, partly serpentinized cumulus olivine (ol) grain. Grains in core of olivine (arrowed) are significantly smaller than those at margin, implying simultaneous crystallization of olivine and chromite. S-poor sample J2-6. B. Chromite grains (s) in partly serpentinized olivine (ol) and anhedral intercumulus ilmenite (T) at boundary between olivine and clinopyroxene. Sample J1-1, disseminated ore. C. Chromite (s) in cumulus olivine (ol), and coarser chromite in intercumulus clinopyroxene (cpx). Sample J1-2, S-poor sample from disseminated ore zone. D, E, and F. Coarse Fe-rich Cr-spinels (s) in matrix ores associated with partly serpentinized olivine (ol), pyrrhotite (po), and pentlandite (pn). Subhedral spinels in (E) show fine rims of ilmenite. Note small euhedral spinels in olivines compared with much more coarse spinels enclosed in sulfide. All reflected light photomicrographs. Sample J2-13.

In net-textured ore, relatively coarse euhedral chrome spinels up to 300 μm across occur enclosed within sulfides or at sulfide-olivine contacts (Fig. 2D, E, F). Coarse, anhedral spinels characterized by high Ti contents and rims of ilmenite also occur in this environment (Fig. 2E, F), and in some cases show very fine lamellar exsolution of a Ti-rich phase, presumed to be ulvospinel (Fig. 3A, B), and intergrowths with sulfides (Fig. 3D). Ilmenite occurs as a rare interstitial phase in olivine cumulates (Fig. 2B), and as exsolved grains from Ti-magnetite in the marginal websterite. It also occurs as thin

rims on some Ti-rich chromites (Fig. 2E), where it is rarely associated with minor amounts of hematite.

Chrome-bearing magnetite, essentially free of Al, occurs as the outer zone of composite zoned chromite-magnetite grains in a few samples. They are restricted to the sulfide-poor surface traverse (Fig. 3C). This magnetite (or "ferritchromite") forms as a result of the reaction of chromite with metamorphic fluids in equilibrium with chlorite (Wylie et al., 1987), and provides evidence that the Jinchuan chromites have, in some cases, been modified by metamorphism. These composite

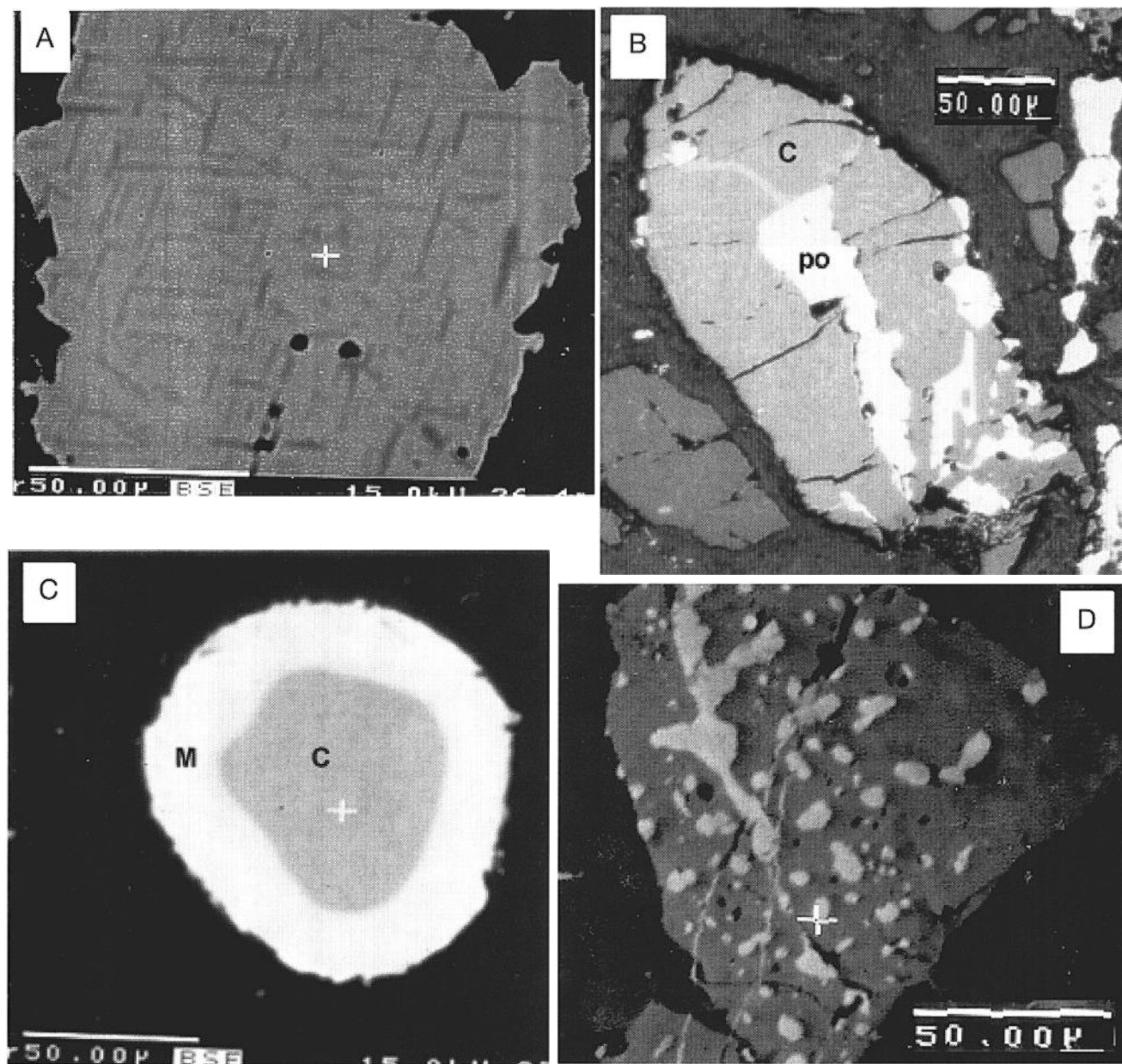


FIG. 3. Details of spinel textures in Jinchuan rocks. A. Lamellar exsolution of Ti-rich phase, probably ulvospinel, from Ti-rich chromite in disseminated ore sample J2-3. B. Ti-rich chromite from matrix ore sample J2-14, with very fine-scale, irregular exsolution of Ti-rich phase, probably ilmenite. C. Complex intergrowth of chromite (C) with pyrrhotite (po). C. Chromite (C) showing partial replacement by magnetite (M) from sulfide-poor sample within unmineralized part of intrusion, sample J2-3. D. Complex graphic intergrowth of chromite with pyrrhotite from matrix ore sample J2-12. All backscattered electron micrographs.

zoned spinels are absent or are at least very rare in ore-zone rocks. Owing to the additional complexity introduced by reaction between chromite, magnetite, and silicates during metamorphic processes (Wylie et al., 1987), composite zoned spinels are excluded from the data presentation and discussion in this paper.

Pure magnetite, as opposed to Cr- and Al-bearing magnetite, occurs as a secondary low-temperature phase generated during serpentinization of olivine. It occurs as micron-scale veinlets within olivine pseudomorphs.

Yejili

The Yejili mafic-ultramafic intrusion, 100 km northeast of Jinchuan City (Fig. 1), was intruded into Proterozoic migmatites and granitoids. It is a north-northeast-elongated body 6,500 m long and 800 to 2,300 m wide, that dips east in the shape of an irregular lens. The intrusion consists mainly of plagioclase peridotite with a discontinuous margin of clinopyroxene peridotite. Samples for this study came from both rock types, while spinels were restricted to olivine orthocumulates with plagioclase oikocrysts and interstitial hornblende.

Olivine shows a very similar style and degree of partial serpentinization as it does in the Jinchuan samples. No economic mineralization has been located in the intrusion. A Nd-Sm isochron has given an age of 773 ± 11 Ma. (Tang, unpub. data).

Oxide minerals: Chromite is the dominant oxide phase, occurring as euhedral and subhedral grains mostly between 50 and 150 μm (Fig. 4A) but with a range of up to 300 μm . The larger grains tend to have more anhedral and occasionally interstitial morphologies. Small subspherical silicate inclusions are a common feature of chromite grains, particularly those not enclosed in olivine (Fig. 4B). Yejili chromites differ from Jinchuan chromites in that they occur most commonly within intercumulus phases or at grain boundaries between cumulus phases. Jinchuan chromites, however, typically occur enclosed within cumulus olivine grains. Yejili chromites commonly occur in clusters surrounded by haloes of late intercumulus hornblende, or associated with intercumulus phlogopite (Fig. 4B). The chromites are homogeneous and lack magnetite rims. Magnetite occurs relatively rarely as a replacement product of disseminated sulfides.

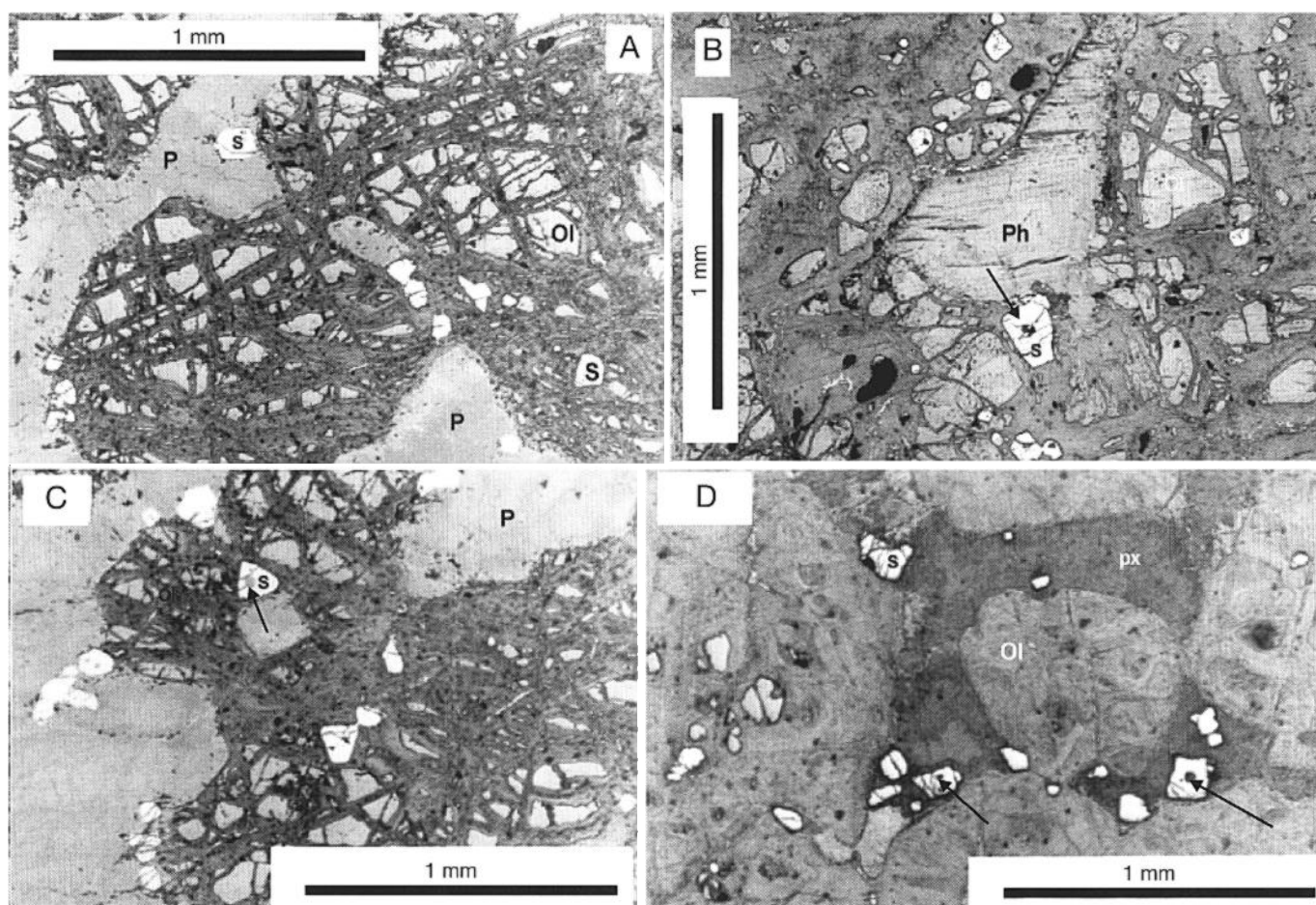


FIG. 4. Spinels in the Yejili and Zangbutai intrusions. A. Chromite (S) at grain boundaries between partially serpentinized olivine (Ol) and plagioclase (P), Yejili sample 89-Y-18. B. Spinels (S) at margins of cumulus olivine (Ol) grains adjacent to intercumulus phlogopite (Ph), Yejili, sample 89-Y-18. Note spherical silicate inclusion in large spinel in lower center. C. Chromite (S) at plagioclase (P)-olivine grain boundaries in Zangbutai sample 89-Zh-10. D. Chromite (S) at olivine (Ol)-pyroxene (px) grain boundaries, Zangbutai sample 89-Zh-10. Note presence of spherical silicate inclusions.

Zangbutai

Geology: The Zangbutai intrusion was intruded into early Proterozoic carbonaceous phyllite and chlorite-quartz schist, about 130 km northwest of the Jinchuan intrusion (Fig. 1). It is a sheetlike, northwest-elongated body of ultramafic rocks 2,300 m long and 120 to 400 m wide, dipping 60° to 80° to the west. The intrusion is composed of clinopyroxene peridotite and amphibole clinopyroxenite with mutually gradational contacts, intruded by gabbro-dolerite dikes. The clinopyroxene peridotite occurs in the middle and lower parts of the intrusion, and the amphibole clinopyroxenite occurs in the margins and upper part. Samples for this study came from both rock types, but spinels were only located in the peridotite. Olivine is almost completely serpentinized, and only rare, relict igneous olivine is preserved. An Nd-Sm isotope study produced an isochron age of 1511 ± 168 Ma, which is similar to that determined for Jinchuan, and the intrusion is believed to be comagmatic with Jinchuan (Tang and Li, 1995). No magmatic sulfide mineralization is known.

Oxide minerals: Chromite occurs mainly as 50- to 150- μ m subhedral grains at olivine grain boundaries, rarely within intercumulus pyroxenes and very rarely entirely enclosed within olivine grains (Fig. 4C, D). Spherical silicate inclusions in chromites are fairly common. Chromites commonly show rimming and partial replacement by Cr-free magnetite.

Spinel Chemistry

Data were collected from 23 samples, including five from Yeji and four from Zangbutai. A total of 465 chromite analyses were obtained, along with 114 silicate analyses (mostly of olivine) and a few analyses of ilmenite. The apparent size of the grain and the enclosing silicate mineralogy was noted for all of the chromites analyzed. Full analytical data are available on the World Wide Web (see Appendix).

The triangular Cr-Al-Fe³⁺ plot (Fig. 5) and plots of major and minor elements, versus $Mg/(Mg + Fe^{2+})$ in Figure 6 and $Fe^{3+}/(Cr + Al + Fe^{3+})$ in Figure 7, compare chromite analyses from the various regions of the Jinchuan intrusion with samples from Yeji and Zangbutai. The data are superimposed, where appropriate data are available, on fields for spinels from layered tholeiitic intrusions based on the extensive database compilation of Roeder (1994). The field boundaries are in percentiles based on data density on the plots. Data from subvolcanic intrusions related to the Karoo flood basalt province in South Africa, and data from the Sudbury intrusive complex, are shown for comparative purposes and are discussed further below.

The data that are plotted represent compositions of individual chromite cores. Composite zoned chromite-magnetite grains from Jinchuan were excluded, owing to the possibility of these core compositions having been modified by diffusive exchange with magnetite and chlorite during metamorphism.

Most of the Jinchuan data points, excluding those from the ore zone, fall within the typical range for layered intrusion spinels and have distinctly lower Cr contents than komatiitic chromites (Fig. 5). This is consistent with the hypothesis of Chai and Naldrett (1992a, b) that the Jinchuan parent magma was a mafic rather than an ultramafic melt. Fe³⁺ contents are negatively related to $Mg/(Mg + Fe^{2+})$, as is typical in layered

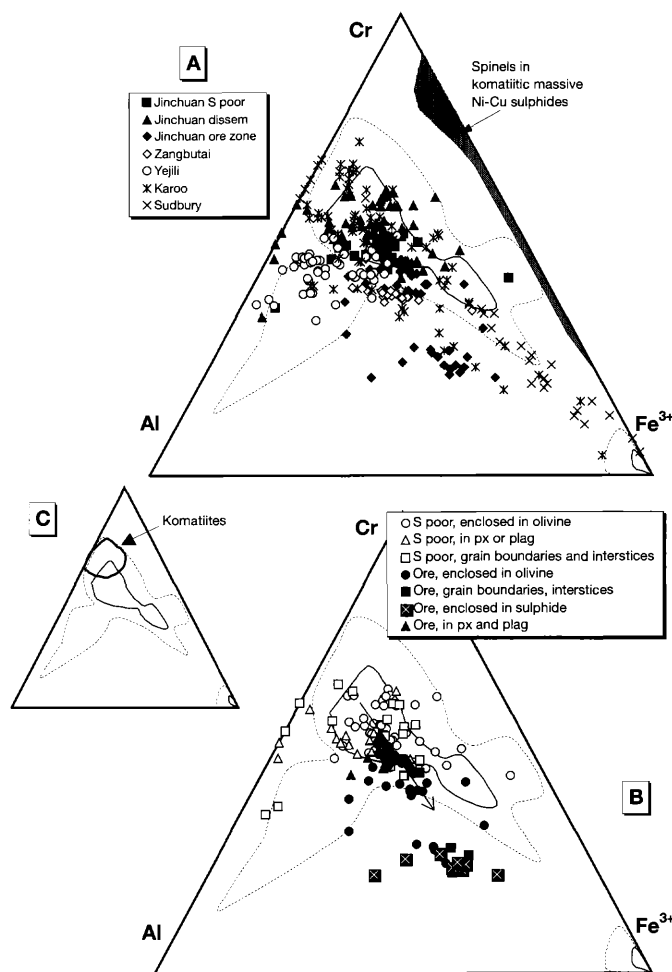


FIG. 5. Trivalent cation plot for spinels from the Jinchuan, Yeji, and Zangbutai intrusions. Chromites with marginal magnetite replacement and with Ti-magnetite exsolution are excluded. Outlines show field for spinels from tholeiitic layered intrusions, from database of Roeder (1994)—dashed line encloses 90 percent of the data, solid line encloses 50 percent of the data. A. Gansu province intrusions compared with data for Karoo intrusions (Eales, 1979; Eales and Snowden, 1979; Lightfoot and Naldrett, 1983; Cawthorn et al., 1991; Cawthorn and Biggar, 1993) and Sudbury irruptive (Zhou et al., 1997), and the field for spinels within komatiite-hosted magmatic massive Fe-Ni sulfide deposits (Groves et al., 1977, and unpub. data). B. Data from Jinchuan intrusion, subdivided by host rock (disseminated sulfide and barren samples lumped together as "S poor" for this plot), and by location of grain in relation to silicate or sulfide minerals. "In sulfide" indicates partially or completely enclosed in sulfide. "Grain boundary or interstices" indicates location at silicate-silicate grain boundaries regardless of phase. "Enclosed in olivine" indicates grain is entirely enclosed within a single olivine grain. Arrow shows direction of trapped liquid trend identified in Kilauea Iki lava lake samples by Scowen et al. (1991). C. 90th and 50th percentiles for layered intrusion spinels, compared with 90th percentile for primary komatiitic chromites from Barnes (1998). Error bars on this and subsequent plots are based on 2σ precisions of replicate analyses of an internal standard, given by Barnes (1998).

intrusions (Fig. 6B). TiO₂ falls within the typical layered intrusion range in most cases, although TiO₂ contents in Jinchuan samples vary widely even on the scale of individual thin sections (Fig. 8). For example, sample J1-2 contains almost the complete range of TiO₂ contents seen in the entire Jinchuan data set. For the Yeji and Zangbutai spinels, and

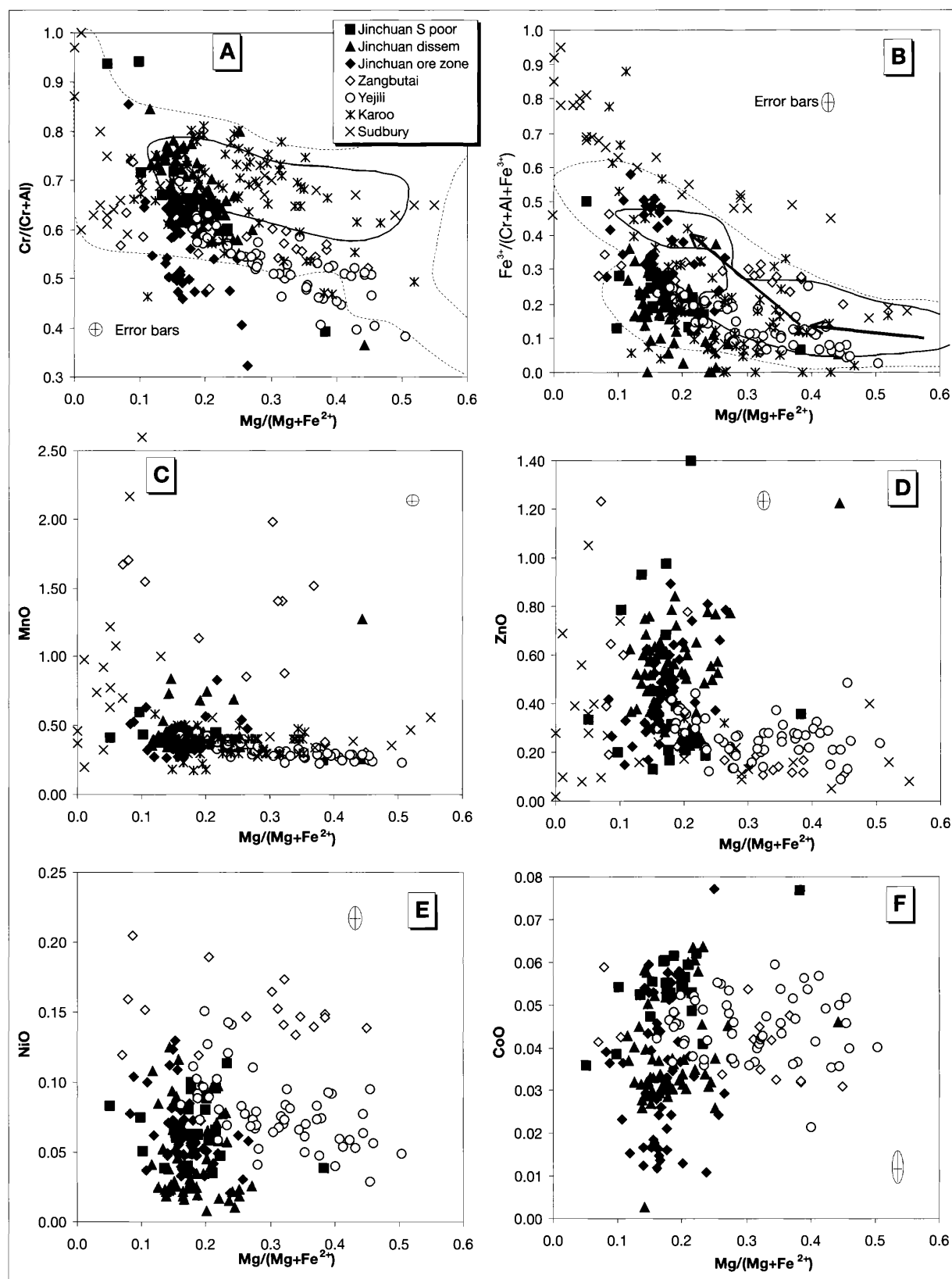


FIG. 6. Divalent cation oxide percentages and molar ratios for trivalent ions for spinels from the Jinchuan, Yejili, and Zangbutai intrusions, plotted against $Mg/(Mg + Fe^{2+})$ and compared with fields for literature data from tholeiitic layered intrusions, where reliable, as in Figure 5. Arrow on $Fe^{3+}/(Cr + Al + Fe^{3+})$ plot shows direction of trapped liquid trend identified in Kilauea Iki lava lake samples by Scowen et al. (1991).

for most of the Jinchuan data, the pattern of variation of TiO_2 with Fe^{3+} is typical of that in layered intrusions—the range of TiO_2 contents widens with increasing Fe^{3+} (Fig. 7). Compared with the other Gansu intrusions, Jinchuan spinels are distinct in ranging to very high TiO_2 contents. Both Yejili and Zangbutai spinels show positive correlations between TiO_2 and Fe^{3+} , similar to those observed in chromites from Kilauea Iki lava lake cumulates (Scowen et al., 1991). This feature is interpreted as the result of the interaction of chromites with crystallizing trapped intercumulus liquid, as discussed further below.

A number of grains, mostly from weakly mineralized Jinchuan samples, show anomalously high TiO_2 at low ferric iron contents (Fig. 7). They fall well outside of the typical layered intrusion range, but well within the range defined by

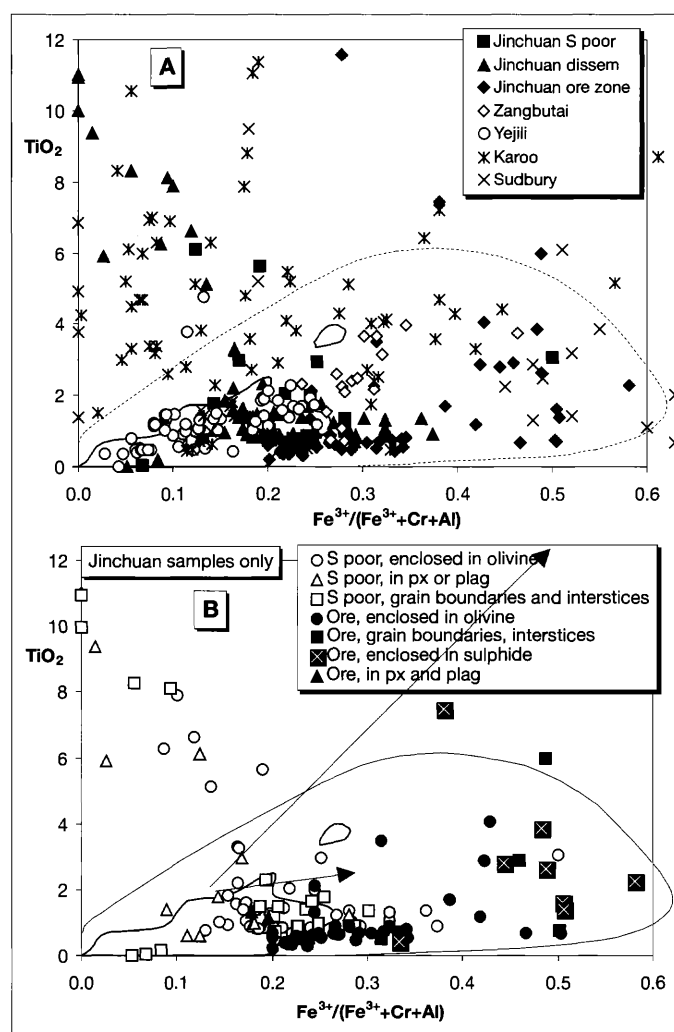


FIG. 7. Plot of $\text{Fe}^{3+}/(\text{Cr} + \text{Al} + \text{Fe}^{3+})$ versus TiO_2 for spinels from the Jinchuan, Yejili, and Zangbutai intrusions, compared with fields for literature data from tholeiitic layered intrusions, as in Fig. 5. A. Gansu province intrusions compared with data for Karoo intrusions and Sudbury eruptive, as in Fig. 5. B. Data from Jinchuan intrusion, subdivided by rock type mode of occurrence of chromite grains as in Fig. 5B. Arrows show direction of Kilauea Iki trapped liquid reaction trends—steeper arrow shows trend for chromites not enclosed in olivines, which extends to $\text{Fe}^{3+}/(\text{Cr} + \text{Al} + \text{Fe}^{3+})$ about 0.5 and 13 percent TiO_2 . Lower arrow shows trend for enclosed chromites.

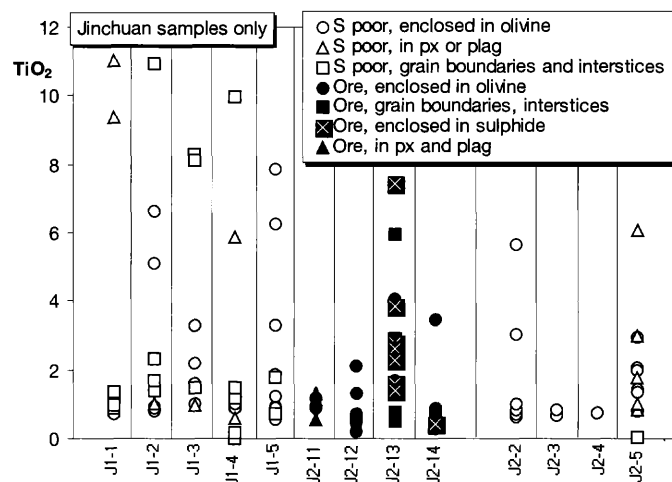


FIG. 8. TiO_2 in Jinchuan spinels in individual samples. Symbols indicate mode of occurrence of grain.

spinel from subvolcanic intrusions in the Karoo flood basalt province in South Africa, as discussed below. Those grains that contain visible Ti magnetite or ulvospinel exsolution are not plotted, as they presumably originally had much higher Ti contents than those now analyzed. Of the grains that are plotted, it is possible that they contain submicroscopic exsolution of ilmenite or ulvospinel. The calculation of ferric iron in spinel rests on the assignment of Fe to Ti in ulvospinel (see Appendix), so that undetected ilmenite exsolution would result in spuriously low calculated values of Fe^{3+} , introducing some uncertainty into the correct positions of these points on the TiO_2 - Fe^{3+} plot. However, unless there is major gain or loss of oxygen during exsolution, this should not have a big effect on the sample positions.

Jinchuan spinels show a stronger resemblance to the Zangbutai data in most other respects, except for Mn (Fig. 6C), for which the Zangbutai samples are strongly enriched relative to both Yejili and Jinchuan. The Yejili and Jinchuan samples show a linear trend of increasing MnO from about 0.2 to 0.4 percent, with decreasing $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ from 0.5 to 0.2 percent, a feature which is also observed in komatiitic chromites (Barnes, 1998) and which is attributable to equilibration with olivine. Elevated Mn is thought to be the result of alteration during serpentinization, on the basis of detailed observation of komatiitic chromites (Barnes, unpub. data), and the high Mn of the Zangbutai samples may reflect the higher degree of serpentinization. The compositions of the Jinchuan samples extend to lower Mg contents and lower Al contents than the Yejili (but not the Zangbutai) samples (Fig. 5A). The Jinchuan samples are also distinctly lower in Ni (for a given $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$) than the other intrusions, and somewhat lower in Co. They are slightly enriched in Zn for a given $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ compared with Yejili, but are similar in Zn contents to Zangbutai.

One group of Jinchuan spinels is anomalously low in Cr and rich in Fe^{3+} , and falls well outside of the typical range of layered intrusion spinels (Fig. 5, lower right center of both triangles, black diamonds on Fig. 5A). These spinels are all from sulfide-bearing net-textured ores from the main ore zone. They occupy an area on the Cr-Al- Fe^{3+} diagram in which

spinel in layered intrusions or in any other environment are rare, and which is within a low-temperature, two-phase field according to the calculations of Sack and Ghiorso (1991). These spinels probably interacted with the sulfide ore magma at high temperatures, but they are compositionally distinct from Fe-rich chromites found associated with massive Ni-Cu sulfide ores in komatiites (Groves et al., 1977), which are typically ferrian chromites with little or no Mg and Al (Fig. 5A).

Fe-Mg Exchange and Olivine-Spinel Geothermometry

The Jinchuan spinels have systematically lower $Mg/(Mg + Fe^{2+})$ values than spinels from Yejili and Zangbutai, and this feature requires an explanation. This section addresses the inconsistency in terms of interdiffusion and equilibration of Fe^{2+} and Mg between spinel and olivine during postcumulus and subsolidus cooling.

In all the samples studied, chromite is present in minor abundances relative to olivine, and therefore subsolidus Fe-Mg exchange exerts a major influence on spinel composition (Irvine, 1967; Roeder et al., 1979). The exchange equilibrium favors more Fe-rich chromite with falling temperature. In a rock with a high olivine/chromite ratio, the final $Mg/(Mg + Fe^{2+})$ value of the chromite is a function of the composition of the coexisting olivine and the effective final equilibration temperature, or blocking temperature, which is itself a function of the cooling regime. Small high-level sills and lava flows record blocking temperatures close to the solidus. Large intrusions typically record much lower temperatures, reflecting slow subsolidus cooling which is manifest as low Mg/Fe ratios in spinel. The low $Mg/(Mg + Fe^{2+})$ values of the Jinchuan spinels may reflect its origin as a deeply eroded remnant of an originally much larger intrusion, and the contrast between Jinchuan and the other intrusions may reflect a combination of blocking temperature and coexisting olivine composition.

Geothermometric calculations were carried out on coexisting spinel-olivine pairs from the Jinchuan and Yejili samples, using the formulation of Sack and Ghiorso (1991). Insufficient fresh olivine is preserved in the Zangbutai samples to allow for the analysis of spinel and olivine in direct contact. Analyses were carried out on points about 20 to 50 μm from the mutual grain boundary, which is about as close as practically possible given that olivines in both intrusions are commonly serpentinized along fractures and grain boundaries. Olivine compositions were found to be homogeneous more than 20 μm away from chromite grain boundaries, but in some cases showed an increase of about 2 mole percent Fo approaching to within 10 μm of the grain boundary. This small-scale zoning was also noted by Wilson (1982) in the Great Dyke, and was attributed by him to incomplete equilibration close to the blocking temperature.

Jinchuan measurements show a bimodal distribution (Fig. 9), with a main peak between 675° and 700°C and a second peak in the 525° to 550°C range. This second peak corresponds to the small number of samples containing zoned spinels with magnetite rims, and is very probably the result of metamorphic reequilibration in these samples. If these measurements are excluded, the Jinchuan data have a median of 688°C and a standard deviation of 37°C over 38 determinations on eight samples. The Yejili samples record a slightly higher median temperature of $711 \pm 27^\circ C$ (one standard deviation) over 13 determinations on four samples.

The Jinchuan temperatures are in broad agreement with those inferred by Wilson (1982) for the Great Dyke. These data do not provide any conclusive indication of cooling rates. However, on the evidence of geothermometry and small-scale zoning of olivine near spinel grain boundaries, the Jinchuan intrusion appears to have cooled at a rate more appropriate to a large, layered intrusion than to a small, high-level intrusion of the present dimensions of the body.

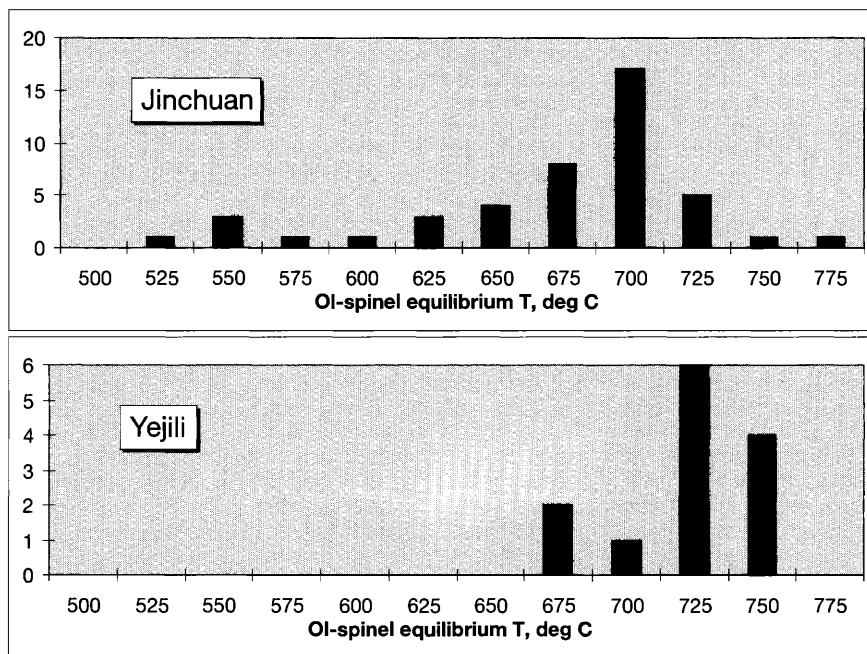


FIG. 9. Distribution of olivine-spinel equilibration temperatures in the Jinchuan and Yejili samples.

The Yejili intrusion has slightly more magnesian olivines on average than Jinchuan (Fig. 10). The substantial difference in $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ values between Jinchuan and Yejili chromites is a consequence of this small difference in the coexisting olivines and a slightly higher effective blocking temperature at Yejili. The higher Al content of the Yejili chromites also tends to lower the K_d for Fe/Mg in chromite at a given temperature (Sack and Ghiorso, 1991). In an olivine-buffered system such as exists in both of these intrusions, a small variation in these variables produces a large effect on $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ in chromite.

Proton microprobe (PIXE) data—gallium and zirconium contents

Chromites in a small group of samples, two from each intrusion, was analyzed on the PIXE (proton induced X-ray excitation) microprobe line of the Heavy Ion Analytical Facility (HIAF) at the CSIRO Division of Exploration and Mining in Sydney. The only trace elements detected in these chromites, other than those already reported, were gallium (Ga) and zirconium (Zr).

The results are not particularly informative. The Yejili chromites are consistently higher in Ga than those in the other two intrusions, but this is to be expected, in that these chromites are also higher in Al, and Ga is known to substitute for Al in most aluminous minerals. There are no systematic differences in Zr between the intrusions, and the data show a wide scatter at very low concentrations close to the detection limit.

Spinel chemistry in relation to enclosing mineralogy

Numerous studies of spinels in cumulate rocks have shown that their compositions can be significantly modified by reaction with evolving trapped intercumulus liquid that is undergoing in situ crystallization. This modification is sensitive to the microenvironment of individual grains. Reaction leads to progressive enrichment in components such as Fe^{2+} , Fe^{3+} ,

and TiO_2 (Henderson, 1975; Henderson and Wood, 1981; Roeder and Campbell, 1985; Scowen et al., 1991; Barnes, 1998). This effect underpins the observed trend of increasing Fe^{3+} and TiO_2 with decreasing Mg number observed in layered intrusions globally (Fig. 6B), and it is essential that it be taken into account when interpreting the data presented here.

Scowen et al. (1991) showed, in a study of the Kilauea lava lake, that chromites could easily exchange Mg and Fe^{2+} by interdiffusion through olivine grains, but that chromites enclosed in early crystallizing olivine were protected from Ti enrichment owing to the difficulty of diffusing Ti through the olivine structure. Ti could exchange, but at a much slower rate. Chromites outside of olivines had TiO_2 contents commonly as high as 8 percent and rarely as high as 13 percent, compared with starting values of between 1 and 2 percent. Spinel from Zangbutai and Yejili are typically not enclosed in olivine, and show the expected Ti- Fe^{3+} enrichment similar to the Kilauea trends.

The Jinchuan data set is more complex. With the exception of the very strongly Ti-enriched spinels with low Fe^{3+} , the data show a widening envelope of increasing Ti with increasing Fe^{3+} , superficially consistent with the Kilauea Iki trend. However, it is evident from Figure 6B that there is no systematic relationship of spinel chemistry to enclosing mineralogy in the Jinchuan samples, apart from a group of low Ti spinels in olivines in matrix ore samples, and some of the most Ti-enriched spinels are enclosed in olivines. The distinctive group of Fe^{3+} -enriched, moderately Ti-enriched spinels is associated with sulfide-rich matrix ores from the Jinchuan ore zone. This group includes spinel grains partly or entirely enclosed within sulfides, but also a few grains that are entirely enclosed in olivine.

Roeder and Campbell (1985) showed that chromites in the Jimberlana intrusion were able to become Ti-enriched even when enclosed in olivine, and postulated that chromites were able to interact with intercumulus melts percolating along

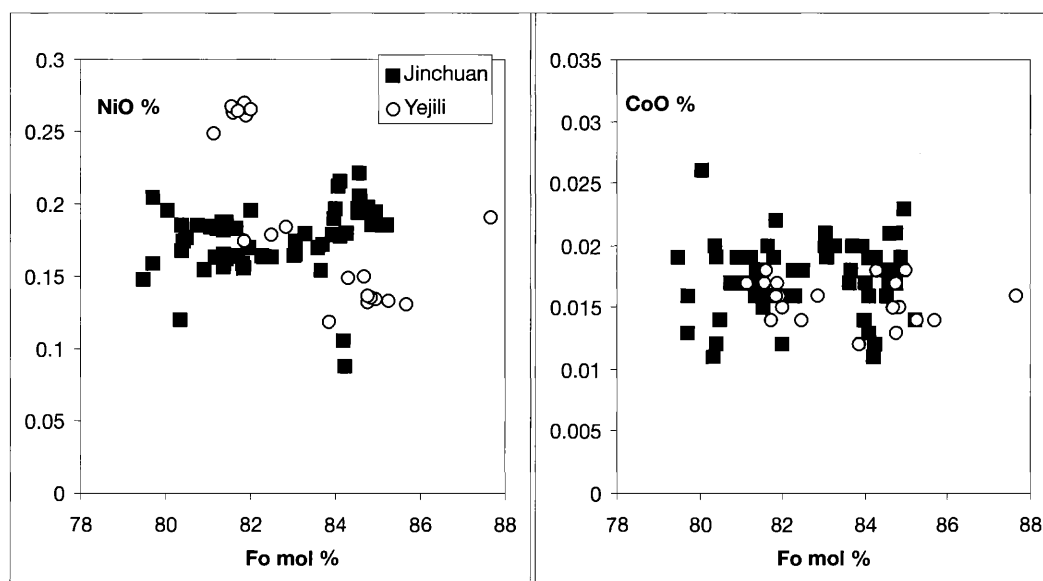


FIG. 10. Comparison of NiO and CoO contents of olivines from the Jinchuan and Yejili intrusions.

microfractures through olivine grains. The same process appears to have operated at Jinchuan. The contrast with the lava lake samples may be a function of a slower cooling environment in intrusions.

Spinel chemistry in relation to sulfide abundance

An unusual group of spinels is characterized by high Fe^{3+} , normal Al, and low Cr contents, and is confined to sulfide-rich ores (Fig. 5B). These grains plot along the trapped liquid modification trend with respect to TiO_2 and Fe^{3+} , but are enriched in Fe^{3+} by a factor of two relative to normal Jinchuan spinels (Fig. 6). This enrichment is not accompanied by unusually low $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ values, but the ratio is probably buffered by subsolidus reaction with olivine (see above). They are outside the 90th percentile field for layered intrusion spinels, falling to the Al-rich side of the normal Fe^{3+} -enrichment trend. It is suspected that the unusual chemistry is the result of interaction with the sulfide liquid; however, the mechanism is not a simple one. The compositions of these spinels is distinctly different from those of ferrian chromites associated with massive magmatic sulfide ores (Fig. 5), and cannot simply be interpreted as spinels that have crystallized directly from the sulfide melt. This is borne out by the fact that some of the grains in this group are entirely enclosed in olivines. It is more likely that these grains crystallized initially from the silicate melt having originally normal chemistry, but were subsequently modified by processes related to crystallization of the sulfide liquid.

On formation, magmatic sulfide liquid contains a substantial proportion of dissolved oxygen, which rapidly exsolves as monosulfide solid solution crystallizes from the sulfide melt, driving up its oxygen fugacity (Naldrett, 1969). The normal response to this is crystallization of magnetite from the sulfide melt. In the present case, it could result in reaction with enclosed chromite grains to form magnetite overgrowths, and high-temperature reequilibration between chromite core and magnetite rims to form effectively more oxidized spinel compositions. Alternatively, and more consistent with the present unzoned nature of these spinels, this could be a continuous single-stage process of growth and high-temperature diffusive reequilibration between original, normal chromite cores and newly grown margins. The process was apparently accompanied by some loss of Cr or gain of Al to produce the observed compositions, possibly due to changing Cr/Al melt-chromite distribution coefficients with changing oxidation state. The coarser grain size of the sulfide-hosted chromites (150–250 μm , compared with the typical 50–150 μm) supports a model involving additional growth of spinel.

The ore zone also contains unusually low Ti chromites with moderate Fe^{3+} contents, more typical of normal layered-intrusion chromites. Most of these occur enclosed within olivine. They can be explained partly by the shielding effect of the interstitial sulfide liquid, preventing these chromites from coming into contact with silicate melt, and also by olivine, inhibiting additional spinel growth and reaction.

The Jinchuan data suggest that chromite reaction trends followed different paths according to the extent of reaction with sulfide. In most cases, regardless of sulfide content, chromites followed typical reaction trends with trapped liquid, becoming enriched in Ti and Fe^{3+} . In sulfide-rich

samples, the same process operated initially, but a proportion of spinel grains subsequently became modified at a late stage owing to reaction with the sulfide melt. The efficacy of both processes depended on whether grains were directly exposed to sulfide liquid or were effectively bottled in unfractured olivine grains.

Chromite grains that are closely intergrown with sulfides are generally Ti enriched relative to the majority of chromites from the Jinchuan ore zone (Fig. 7B). This is not what would be expected if these rocks formed in a single stage as olivine-sulfide-chromite cumulates, since Ti is largely insoluble in sulfide liquid. Either these Ti-rich grains have primary liquidus compositions, markedly different from the majority of grains in the same rock, or they had a prehistory of extensive reaction with trapped silicate liquid before being surrounded by sulfide liquid. This implies that sulfide liquid may have been physically mobile during solidification of the cumulate pile and displaced interstitial silicate liquid. In some cases, spinel growth continued during the displacement process, giving rise to complex spinel-sulfide intergrowths such as those shown in Figure 3B and D.

The possibility that Ti-rich spinels are primary in origin, reflecting liquidus compositions derived from some anomalously Ti parent magma, was suggested by Cawthorn et al. (1991) for similar spinels from the Mt Ayliff (or Insizwa) intrusion in South Africa. This is discussed further below.

Ni and Co depletion in Jinchuan chromites

The Jinchuan spinels are systematically depleted in Ni and Co relative to the barren intrusions. This is true for both S-poor and S-rich samples, but more so for S-rich samples (Figs. 6E, 11). The depletion in Ni is more apparent when covariance with Fe^{3+} is taken into account.

Nickel contents of spinels from the three intrusions studied show positive correlations between NiO and $\text{Fe}^{3+}/(\text{Cr} + \text{Al} + \text{Fe}^{3+})$, when two groups of data are considered: the barren Yejili and Zangbutai intrusions as one group, and the S-poor rocks of the mineralized Jinchuan intrusion as the other. There is no Ni- Fe^{3+} correlation for the Jinchuan ore zone group. Positive correlation of Ni with Fe^{3+} has been noted previously in chromites in komatiites at low metamorphic grades in the absence of extensive magnetite replacement (Barnes, 1998). This relationship probably reflects the strong octahedral site preference energy of the Ni^{2+} ion, combined with the tendency for spinels to adopt the inverse spinel structure as they become richer in Fe^{3+} (Lindsley, 1981). This results in increased affinity of Ni^{2+} for Fe^{3+} -rich spinels.

The variation of Ni with Fe^{3+} in the Yejili, Zangbutai, and S-poor Jinchuan groups is interpreted as a further consequence of trapped liquid equilibration. The partition coefficient of Ni from silicate melt into spinel increases during crystallization due to reduction of octahedral sites in the melt with falling temperature and evolving composition, and evolution toward more Fe^{3+} -rich chromites. The absence of the correlation in Jinchuan S-rich samples and the strong depletion in Ni in chromites in these samples imply that competition for Ni from coexisting sulfide liquid produces a competing effect.

Roeder (pers. commun., 1998) interprets Ni variations in chromite in terms of release of Ni from serpentinized olivine.

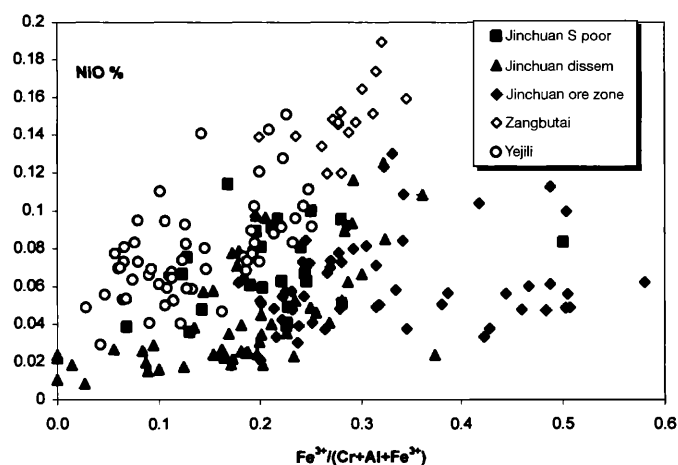


FIG. 11. Plot of $\text{Fe}^{3+}/(\text{Cr} + \text{Al} + \text{Fe}^{3+})$ versus NiO for spinel cores from the Jinchuan, Zangbutai, and Yejili intrusions.

There are several reasons why we feel that this is not the case. The host rocks at Jinchuan and Yejili contain very similar olivines with similar degrees of serpentinization (Figs. 1, 3, 8), but Ni contents of chromites are distinctly different. Zangbutai samples show much higher degrees of serpentinization, but plot on the same linear trend as Yejili. Fe and Mg in the Yejili and Jinchuan chromites have been shown to have equilibrated with olivine at temperatures above 750°C , well above the maximum stability of serpentine. If Ni had reequilibrated at much lower temperatures, then Fe and Mg should have done the same. Barnes (1998) presented an extensive data set on Ni in komatiitic chromites in which Ni shows distinctive patterns of variation in different crystallization environments, but shows no relationship to degree of serpentinization. For these and other reasons the variation seen here is believed to be a genuine magmatic effect.

Jinchuan chromites also show evidence of Co as well as Ni depletion (Fig. 6F) relative to the barren intrusions, although there is more overlap than in the case of Ni. The strongest Co depletion is seen in the sulfide-associated group, and to a lesser extent in most of the disseminated ore data.

There are two possible explanations for the relative depletion of Ni and Co in Jinchuan chromites. First, the Jinchuan parent magma could have been Ni and Co depleted as a result of sulfide liquid segregation. Second, the chromites could be Ni and Co depleted as a result of interaction with cumulus sulfide liquid during the process of reaction with trapped liquid.

If the first hypothesis is correct, then Ni and Co depletion in the parent magma should also be reflected in olivine. Jinchuan olivines should also be Ni and Co depleted relative to the barren intrusions, which they are not (Fig. 10). Both Jinchuan and Yejili olivines span a wide range in Ni and Co, with no systematic relationship to forsterite content of olivine, but there is clearly no discernible depletion in the Jinchuan samples.

Ni and Co depletion during reaction with trapped liquid, as a result of competition for both elements with the coexisting sulfide liquid, is therefore the most likely explanation. This is consistent with the fact that the most Ni- and Co-depleted chromites are in the most sulfide-rich samples. The exchange

of Ni, Co, and Fe^{2+} between chromite and sulfide at the prevailing equilibration temperatures apparently favors partitioning of Ni and Co from chromite into sulfide. Regardless of the precise mechanism, there is a clear contrast between spinels from mineralized and barren environments.

Comparison with Intrusions Related to Flood Basalts and with the Sudbury Irruptive, and the Origin of Ti-rich Chromites

Titanium enrichment is a distinctive feature of some Jinchuan chromites. Titanium-rich chromites have been reported from a number of intrusions associated with continental flood basalt provinces, particularly the Insizwa and Elephant's Head intrusions of the Karoo province (Eales, 1979; Eales and Snowden, 1979; Lightfoot and Naldrett, 1983; Cawthorn et al., 1991; Cawthorn and Biggar, 1993), and more rarely in flood basalts themselves (Thompson, 1973). Strong Ti enrichment has also been reported in spinels, mainly magnetites but also some chromites, from mafic and ultramafic inclusions in the Sudbury sublayer (Zhou et al., 1997). These data are compared with the Jinchuan data in Figures 5, 6, and 7.

The Karoo data overlap with typical layered intrusion compositions, and show a broadly similar range in TiO_2 to the Jinchuan spinels (Fig. 7). The Jinchuan data set contains a much higher proportion of low-Ti spinels, attributable to partial shielding by olivine. Overall, the Jinchuan data show significantly higher Al contents than the Karoo spinels, and lower $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ for a given TiO_2 . The latter may be due to an average lower temperature of final equilibration of Jinchuan spinels with olivine, as discussed above. Yejili and Zangbutai show a similar range of $\text{Mg}/(\text{Mg} + \text{Fe}^{2+})$ to the Karoo spinels, but are substantially more aluminous.

In summary, the overlap in TiO_2 contents suggests that the parent magma to Jinchuan evolved along a similar path to the Karoo magmas. The significantly higher Al contents of the Jinchuan and Yejili spinels is probably a consequence of parent magma composition, and may be related to the delayed onset of plagioclase crystallization in these intrusions relative to the Karoo intrusions.

The interpretation favored here is that strong Ti enrichment in a small proportion of Jinchuan spinels is related to localized, very extensive reaction with highly fractionated, trapped intercumulus silicate melt. The other possible interpretation is that they are primary in origin and reflect liquidus compositions derived from some anomalously high Ti parent magma, as suggested by Cawthorn et al. (1991) for similar spinels from the Mt Ayliff (or Insizwa) intrusion in South Africa. Both interpretations have problems.

A primary liquidus origin is difficult to reconcile with the preponderance of normal Jinchuan chromites with TiO_2 contents of 1 percent or less, typical of layered intrusions, implying crystallization from fairly typical parent magmas and coexisting on a scale of millimeters with high-Ti spinels (Fig. 8). A very unusual, highly Ti-rich magma must somehow have been present in the intrusion, intimately associated with a normal low-Ti magma, but capable of leaving a record of unusual spinels that survived interaction with the normal magma. The intimate coexistence of low- and high-Ti

chromites on thin section scale is good evidence for the origin of Ti enrichment by grain-scale, trapped liquid processes.

On the other hand, if trapped liquid equilibration is the correct explanation, highly Ti-rich spinels should be relatively common in tholeiitic layered intrusions. Reported occurrences are rare, but not unknown: just over 5 percent of the 900 layered intrusion samples reported in Roeder's 1994 database have TiO_2 in excess of 3 percent. There is also reason to believe that high-Ti spinels may be under-represented in this data set. All the high-Ti spinels reported occur as very minor phases in the host rock, a situation in which they are most likely to be modified by trapped liquid reaction. Most of the published literature on layered intrusion chromites is on relatively chromite-rich rocks, including a high proportion of the literature on chromite seams, where the ratio of chromite to trapped liquid in the rock is very high. In this situation, cumulus chromite would itself be the major repository of Ti in the rock, and trapped liquid reaction would have a minimal effect. Accessory chromites in normal, chromite-poor orthocumulates are probably under-represented in the layered intrusion data set. However, the fact remains that they are rare.

Ti enrichment is possible in cumulates derived from Ti-poor magmas, or at least from magmas which were in no way anomalously Ti rich. TiO_2 contents of more than 2.5 percent have been found in slowly cooled olivine orthocumulates derived from primitive komatiite magma (Barnes, 1998). Furthermore, the degrees of enrichment required for the Jinchuan spinels is comparable to that found by Scowen et al. (1991) in the Kilauea lava lake, where reacted chromites had TiO_2 contents as high as 13 percent, compared with starting values between 1 and 2 percent.

Zhou et al. (1997) reported strongly Ti-enriched spinels from sublayer inclusions and footwall intrusions at Sudbury, and their data are compared with data from this study and from Karoo intrusions in Figure 5. The compositional trends they report are similar to those observed in the Karoo samples and in interstitial spinels from Jinchuan, and match trends observed in Cr-rich magnetites in typical layered intrusions. The Sudbury data set also includes rare true chromites with high TiO_2 , up to 9 percent, and low Fe^{3+} contents similar to those reported here and in the Karoo intrusions. Strong Ti enrichment appears to be a consequence of prolonged trapped liquid crystallization at Sudbury also, but there is no suggestion that the Sudbury igneous complex has flood basalt associations. These trends are probably a consequence of the cooling regime rather than the initial magma composition.

The origin of the very Ti-rich spinels at Jinchuan with more than 10 percent TiO_2 and low Fe^{3+} remains enigmatic, and clearly more detailed work is necessary to evaluate the occurrence of high-Ti chromite in cumulates. The diversity of Ti-Fe trends at Jinchuan and in the Karoo intrusions may reflect variability in oxygen fugacity trends during solidification. The generally strong similarity between the Jinchuan and Karoo spinels could be consistent with an origin of the Jinchuan intrusion as a feeder to a continental flood basalt province, but the Ti-rich spinels do not provide conclusive evidence and are a relatively small component of a population dominated by normal, low-Ti chromites. The interpretation favored here is

that Ti enrichment is a trapped liquid reaction effect and contains little information about original parent magmas.

Conclusions

Chromites from the Jinchuan intrusion show many features that are typical of normal tholeiitic layered intrusions. Much of the compositional variance can be attributed to reaction between cumulus chromite and trapped intercumulus liquid. This process operated extensively even on grains that were armored by cumulus olivine crystals. Different reaction paths correlate with the abundance of sulfides in the rock.

The low Cr, high Fe^{3+} chromites from the Jinchuan ore zone are unusual and distinctive. A plausible model for these unusual compositions is that these grains originated as normal aluminous spinels crystallizing from the parent silicate melt, and subsequently become enlarged and enriched in magnetite component through reaction with the sulfide melt during crystallization.

Ti-enriched chromites enclosed within sulfides probably record a prehistory of extensive reaction with trapped liquid. Sulfide liquid was physically mobile during the solidification process and displaced intercumulus silicate liquid.

The chromites are also strongly nickel depleted, relative to expected values for their Fe^{3+} content, a feature that is shared with all chromites from sulfide-bearing samples at Jinchuan when compared with the barren intrusions. Nickel depletion is a hallmark of the Jinchuan chromites, and this feature, combined with the distinctive chemistry of the sulfide-related spinels, may have applications in the use of heavy resistate minerals during exploration.

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March 5, 1998; January 23, 1999

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APPENDIX

Electron Microprobe Analytical Methods

All elements were analyzed by wavelength-dispersive electron microprobe analysis using standard procedures on the Cameca SX50 instrument at CSIRO Division of Exploration and Mining, Floreat Park, Western Australia. The elements Si, Ti, Al, Fe, Mg, Mn, Ca, and Zn were analyzed under the following conditions: accelerating voltage 30 Kv, beam current 30 nA, using stepwise peak integration and background stripping, with online ZAF correction procedures from Ware (1981) and Ware et al. (1988). The major elements were counted for 20 seconds, and minor elements Ti, Mn, and Zn for 80 seconds each. Trace elements Ni, Co, and V were analyzed using the CSIROTrace procedure (Robinson and Graham, 1992), under conditions of 30 Kv accelerating voltage and 450 nA, with counting times of 50 to 100 seconds, and offline ZAF correction. A full listing of analytical data is available on the internet at <http://www.per.dem.csiro.au/staff/BarnesSteve/Jinchuan/Jinchuan_abstract.htm> or can be obtained from the senior author on request.

Major elements Cr, Al, Mg, and Fe were standardized on Smithsonian standard magnesiochromite USNM 117075 (Jarosewich et al., 1980), from Tiebaghi mine, New Caledonia (Barnes, 1998, table A2.1). Other elemental standards were as follows: Zn on pure gahnite; Ni, Co, V, Ti, and Mn on pure metals; Si on Smithsonian standard Springwater meteorite forsterite, USNM 2544.

The ferric iron content of each analysis was determined by assuming stoichiometry, and an ideal XY_2O_4 formula where $X = (Fe^{2+}, Ni, Mn, Co, Zn)$ and $Y = (Cr^{3+}, Fe^{3+}, Al)$. Ti was assumed to be present as an ulvospinel component, and V is assigned to a fictive $Fe_7V_2O_{12}$ component. Iron is subdivided into ferrous and ferric iron. The precision of an individual analysis can be assessed from the replicate analyses of the chromite standard presented in table A2.1 of Barnes (1998).